

Peripherals: Working Document

Digital Audio Player • Rev 8

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1. Summary

This document covers the user-interface peripherals: the display, the front navigation inputs, and the side controls. All inputs connect to ESP32-S3 GPIOs; the display connects over SPI.

Peripheral	Qty	Part / Status
Display	1	BuyDisplay ER-TFT024IPS-3: 2.4 in IPS, 240 × 320, ST7789V, 4-wire SPI, no touch
Front switches (up, down, select)	3	C&K PTS810 SJM 250 SMTR LFS, top-actuated SMD
Rotary encoder	1	Alps EC10E1220501, 10 mm hollow-shaft incremental encoder
Side switches (power, volume up, volume down)	3	E-Switch TL1014BF220QG, side-actuated SMD

2. Design Requirements

Requirement	Value
Display	2.4 in IPS, 240 × 320, SPI; ST7789 or ILI9341 driver; glass within ≈ 43 × 60 mm (inside the battery footprint); vendor must publish the FPC pinout and backlight configuration
Front navigation	Three switches (up, down, select) and one rotary encoder on the front face
Side controls	Power, volume up and volume down on the enclosure side
Wake from standby	Any input that wakes the player must use an RTC-capable ESP32-S3 GPIO
Enclosure	≈ 56 × 75 × 13.2 mm; switch and encoder heights must fit
Constraints	Low noise · Small footprint

3. Interface

Peripheral	Connection
Front and side switches	One ESP32-S3 GPIO each, pulled up, switch to GND; debounced in firmware or with an RC filter
Rotary encoder	Mechanical contacts A and B to two ESP32-S3 GPIOs decoded by the pulse counter (PCNT) peripheral, each with a pull-up and an RC noise filter; common terminal to GND. No supply current, so it can act as a wake source.
Display	SPI (SCLK, MOSI, CS, D/C), reset, and a backlight control output

4. External Components

Ref	Part	Notes	Qty
J_{LCD}	Amphenol FCI F32D-1A7Y1-21050	FPC connector for the display: 50 positions, 0.5 mm pitch, right-angle SMT, dual contact (top and bottom), back-flip ZIF, 2.00 mm height, 0.30 mm cable thickness, 30 mating cycles, gold-plated phosphor-bronze contacts, -40 to 105 °C.	1
SW_UP, SW_DN, SW_SEL	C&K PTS810 SJM 250 SMTR LFS	Top-actuated, J-lead SMD; 4.2 × 3.2 mm; 2.5 mm actuator height; 160 gf; 0.15 mm travel; 150,000 cycles; 50 mA at 16 V DC; -40 to 85 °C. The enclosure key presses the actuator.	3
ENC	Alps EC10E1220501	Incremental (A and B), 10 mm hollow shaft, 9.0 mm mounted height, 24 detents, 12 pulses per revolution, 100,000 cycles, rated 1 mA at 5 V DC. The knob is a custom part through the hollow shaft.	1
SW_PWR, SW_VOLU, SW_VOLD	E-Switch TL1014BF220QG	Side-actuated, right-angle SMD; 4.7 × 3.5 mm; 220 gf; 50 mA at 12 V DC. The enclosure button presses the actuator.	3
LCD	BuyDisplay ER-TFT024IPS-3 (no-touch version)	2.4 in IPS TFT, 240 × 320, ST7789V (COG), 300 cd/m ² , 80° viewing all directions. Outline with FPC folded 42.72 × 59.46 × 2.3 mm; active area 36.72 × 48.96 mm. 50-pin, 0.5 mm FPC, 0.30 ± 0.03 mm thick. Used in 4-wire 8-bit serial interface I.	1

5. Display Interface

FPC pin	Signal	Connection
1	LEDA	Backlight common anode (see Open Problems)
2 to 5	LEDK1 to LEDK4	Backlight cathodes, four LEDs in parallel
6 to 9	IM0 to IM3	IM3 IM2 IM1 IM0 = 0 1 1 0: 4-wire 8-bit serial interface I (SCL, SDA, D/CX, CSX)
10	RESET	ESP32-S3 GPIO
36	WRX (D/CX)	D/CX in serial mode, ESP32-S3 GPIO
37	D/CX (SCL)	SPI clock
38	CSX	SPI chip select
39	TE	Tearing-effect output; optional ESP32-S3 input for tear-free updates
40, 41	VDDI	Interface supply, 1.65 to 3.3 V (2.8 V typical)
42	VCI	Logic supply, 2.5 to 3.3 V (2.8 V typical)
43, 48 to 50	GND	GND
34	SDI (SDA)	SPI data, bidirectional in interface I; ESP32-S3 MOSI

FPC pin	Signal	Connection
11 to 14	VSYNC, HSYNC, DOTCLK, DE	Unused (RGB interface): tie to GND
15 to 32	DB17 to DB0	Unused (parallel bus): tie to GND
33	SDO	Unused output: leave open
35	RD	Unused (parallel read strobe): tie to VDDI
44 to 47	Touch	Not connected (no-touch version)

Serial write clock: 16 ns minimum cycle (62.5 MHz); reads 150 ns minimum (6.7 MHz). Unused inputs are fixed at VDDI or DGND; unused outputs (SDO, TE) are left open. Backlight: four LEDs in parallel with a common anode, V_f 3.2 V typical and 3.4 V maximum at 80 mA total (4×20 mA); normal forward current 70 to 80 mA. No boost converter is needed. Logic inputs: $V_{IH} \geq 0.7 VDDI$, $V_{IL} \leq 0.3 VDDI$. Recommended FPC connector per the panel: 50 pin, 0.5 mm, SMD horizontal, top contact; the dual-contact F32D satisfies this.

Source: [ER-TFT024IPS-3 Datasheet \(EastRising / BuyDisplay\)](#), Sections 2, 3.1 and 4.1 to 4.4; ST7789VW controller datasheet v1.0 (Sitronix, project file), Section 6 pin descriptions and Section 7.4.3.

6. Open Problems

#	Problem
1	Backlight drive: V_f is 3.2 V typical and 3.4 V maximum at 80 mA, so a 3.3 V rail leaves too little headroom for full current. Choose the supply, current-setting method and PWM dimming.
2	Display supply: VCI and VDDI allow up to 3.3 V. +3V3_DIG is 3.318 V nominal, 18 mV above the recommended maximum and far below the 4.6 V absolute maximum. Accept it or lower the digital rail.
3	Encoder pull-up and RC filter values, from the EC10E catalog.
4	GPIO assignment, including RTC-capable pins for every wake source.

7. Revision Log

Rev	Date	Change
0	2026-09-27	Initial issue: peripheral set, requirements, interface.
1	2026-09-27	Side switches: E-Switch TL1014BF220QG.
2	2026-09-27	TS126CJ specifications recorded; encoder changed to Grayhill 62V11-02-P.
3	2026-09-27	Front switches changed to three C&K PTS810 (up, down, select); encoder changed to Alps EC10E1220501.
4	2026-09-27	Display selected: BuyDisplay ER-TFT024IPS-3 (ST7789V, 4-wire SPI).
5	2026-09-27	Display connector: Amphenol FCI F32D-1A7Y1-21050.
6	2026-09-27	F32D specifications confirmed from the manufacturer description: 2.00 mm height, 0.30 mm cable, 30 mating cycles.
7	2026-09-27	Panel datasheet values recorded: outline, FPC thickness, interface mode, supplies, backlight. Backlight drive and supply-level notes added.
8	2026-09-27	Serial data pin and unused-pin terminations from the ST7789VW datasheet; SPI clock limits recorded.

8. Sources

1. [ER-TFT024IPS-3 Datasheet \(EastRising / BuyDisplay\)](#)

2. [ER-TFT024IPS-3 Product Page and Datasheet \(BuyDisplay / EastRising\)](#)
3. [F32D-1A7Y1-21050 Product Page and Datasheet \(DigiKey\)](#)
4. [PTS810 SJM 250 SMTR LFS Product Page and Datasheet \(DigiKey\)](#)
5. [EC10E Series Product Page and Catalog \(Alps Alpine\)](#)
6. [EC10E1220501 Product Page \(DigiKey\)](#)
7. [TL1014BF220QG Product Page and Datasheet \(DigiKey\)](#)